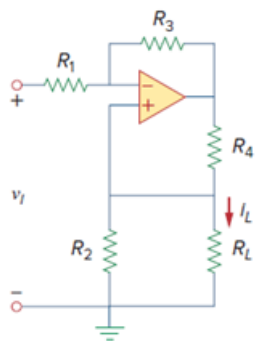


Problem

A voltage-to-current converter is shown in Fig. 5.110, which means that $i_L = Av_i$ if $R_1R_2 = R_3R_4$. Find the constant term A .

Figure 5. 110:



Step-by-step solution

Step 1 of 4

Refer to Figure 5.110 in the textbook.

Draw the circuit as shown in Figure 1.

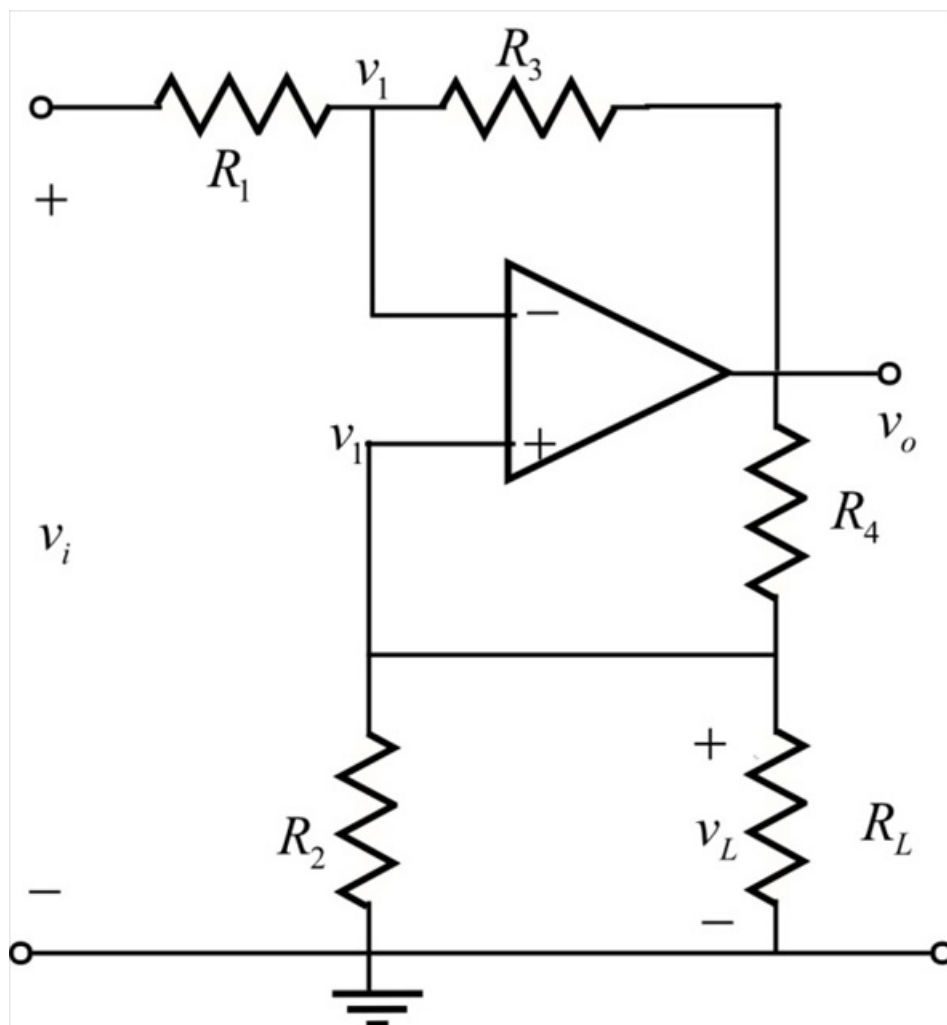


Figure 1

Step 2 of 4

Apply Kirchhoff's current law at node v_1 .

$$\frac{(v_i - v_1)}{R_1} = \frac{(v_1 - v_0)}{R_3}$$

$$(v_i - v_1) = \left(\frac{R_1}{R_3}\right)(v_1 - v_0)$$

$$v_i + \left(\frac{R_1}{R_3}\right)v_0 = \left(1 + \frac{R_1}{R_3}\right)v_1 \dots\dots (1)$$

From the Figure 1,

$$v_1 = v_L$$

Step 3 of 4

Write equation 1 as,

$$v_i = \left(1 + \frac{R_1}{R_3}\right)v_L - \left(\frac{R_1}{R_3}\right)v_0 \dots\dots (2)$$

Consider i_o is the current through R_4 .

$$i_o = \frac{v_0}{R_4 + (R_2 \parallel R_L)}$$

$$i_L = \left(\frac{R_2}{(R_2 + R_L)}\right)i_o$$

$$i_L = \left(\frac{R_2}{(R_2 + R_L)}\right)\left(\frac{V_0}{R_4 + (R_2 \parallel R_L)}\right)$$

$$v_0 = i_L \left[\frac{(R_2 + R_L)[R_4 + (R_2 \parallel R_L)]}{R_2} \right] \dots\dots (3)$$

Step 4 of 4

Consider,

$$v_L = i_L R_L \dots\dots (4)$$

Substituting (3) and (4) into (2).

$$v_i = \left[\left(\frac{(R_3 + R_1)}{R_3} \right) R_L - R_1 \left\{ \frac{(R_2 + R_L)}{(R_2 R_3)} \left(R_4 + \left(\frac{R_2 R_L}{(R_2 + R_L)} \right) \right) \right\} \right] i_L$$

$$\frac{i_L}{v_i} = \frac{1}{\left[\left(\frac{(R_3 + R_1)}{R_3} \right) R_L - R_1 \left\{ \frac{(R_2 + R_L)}{(R_2 R_3)} \left(R_4 + \left(\frac{R_2 R_L}{(R_2 + R_L)} \right) \right) \right\} \right]} \dots\dots (5)$$

Consider the general gain equation for the voltage to current converter.

$$\frac{i_L}{v_i} = A \dots\dots (6)$$

Compare equation 5 with equation 6.

$$A = \frac{1}{\left[\left(\frac{(R_3 + R_1)}{R_3} \right) R_L - R_1 \left\{ \frac{(R_2 + R_L)}{(R_2 R_3)} \left(R_4 + \left(\frac{R_2 R_L}{(R_2 + R_L)} \right) \right) \right\} \right]}$$

Therefore, the gain A is $\boxed{\frac{1}{\left[\left(\frac{(R_3 + R_1)}{R_3} \right) R_L - R_1 \left\{ \frac{(R_2 + R_L)}{(R_2 R_3)} \left(R_4 + \left(\frac{R_2 R_L}{(R_2 + R_L)} \right) \right) \right\} \right]}}$.